ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE E · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Electronics and Digital Concepts for Robotics paper 1 · practice e.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and neat, recognisable diagrams.

SECTION A

QUESTION 1

1.1 C ✓
1.2 A ✓
1.3 D ✓
1.4 B ✓
1.5 A ✓
1.6 B ✓
1.7 D ✓
1.8 C ✓
1.9 A ✓
1.10 B ✓

(10)

[10]

QUESTION 2

2.1 True ✓
2.2 True ✓
2.3 False ✓
2.4 True ✓
2.5 False ✓
2.6 True ✓
2.7 True ✓
2.8 True ✓
2.9 False ✓
2.10 False ✓

(10)

[10]

QUESTION 3

3.1 F ✓
3.2 C ✓
3.3 H ✓
3.4 D ✓
3.5 A ✓

3.6 G ✓

3.7 I ✓

3.8 B ✓

3.9 J ✓

3.10 E ✓

(10)

[10]

TOTAL SECTION A: [30]

SECTION B

QUESTION 4

4.1 Hold the thumb, first finger and second finger of the left hand at right angles to each other ✓: the First finger points in the direction of the magnetic Field ✓, the seCond finger in the direction of the Current, and the thuMb gives the direction of the Motion (force) ✓. (3)

4.2 $\mathrm{F = B \times I \times l \checkmark = 0{,}5 \times 6 \times 0{,}25 \checkmark = 0{,}75\ N \checkmark}$ (3)

4.3 Any FOUR ✓✓✓✓: increase the current; increase the number of turns; use a soft-iron core; make the core larger / shorter gap; wind the turns closer together. (4)

4.4 Current through the coil makes it an electromagnet ✓; it attracts the armature ✓, which moves the contacts to open or close the separate switched circuit ✓. (3)

4.5 Any TWO ✓✓: relays and contactors; solenoid valves; electromagnetic grippers for steel parts; electric locks; motors; electromagnetic brakes. (2)

[15]

QUESTION 5

5.1 $\mathrm{V_2 = V_s \times R_2 \div (R_1 + R_2) \checkmark = 10 \times 3 \div 5 \checkmark = 6\ V \checkmark}$ (3)

5.2 $\mathrm{I = 10 \div 5\,000 \checkmark = 0{,}002\ A\ (2\ mA) \checkmark}$ (2)

5.3 $\mathrm{\eta = 400 \div 500 \times 100 \checkmark = 80\% \checkmark; losses = 500 - 400 = 100\ W \checkmark}$ (3)

5.4 $\mathrm{E = 2 \times 3 \checkmark = 6\ kWh \checkmark; cost = 6 \times 2{,}80 \checkmark = R16{,}80 \checkmark}$ (4)

5.5 When the armature turns in the magnetic field, an emf is induced in it that opposes the supply voltage – the back emf ✓✓. At start-up the armature is not turning, so there is no back emf and only the small armature resistance limits the current ✓, which makes it very large ✓. (4)

5.6 $\mathrm{I = 12 \div (4 + 8) \checkmark = 1\ A \checkmark; V_1 = 1 \times 4 = 4\ V \checkmark; V_2 = 1 \times 8 = 8\ V \checkmark}$ (4)

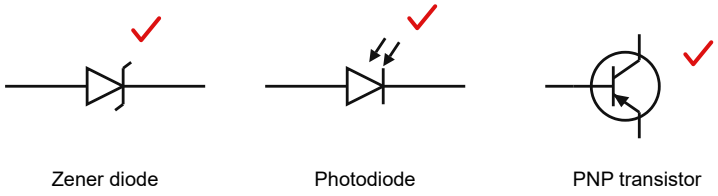
[20]

QUESTION 6

6.1 N-type: pure silicon is doped with a pentavalent impurity (5 valence electrons, e.g. phosphorus or arsenic) ✓, giving extra free electrons ✓. P-type: silicon is doped with a trivalent impurity (3 valence electrons, e.g. boron or gallium) ✓, giving holes ✓. (4)

6.2 An LED gives off light when current flows through it in the forward direction ✓ – e.g. status / indicator light, IR transmitter on a line sensor ✓. A photodiode is used in reverse bias and lets more current flow when light falls on it ✓ – e.g. light sensor, IR receiver, encoder ✓. (4)

6.3



(Zener diode with the bent bar 1, photodiode with the arrows pointing INTO it 1, PNP transistor with the emitter arrow pointing INWARDS 1)

(3)

6.4 Without a resistor the current would be too large and the LED would burn out ✓; $R = (9 - 2) \div 0{,}02$ ✓✓ $= 350\ \Omega$ ✓ (4)

[15]

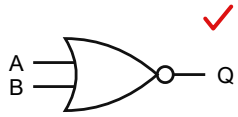
QUESTION 7

7.1.1 $45 = 32 + 8 + 4 + 1$ ✓ $= 101101_2$ ✓ (2)

7.1.2 $16 + 8 + 0 + 2 + 1$ ✓ $= 27$ ✓ (2)

7.1.3 $2 \times 16 + 15$ ✓ $= 47$ ✓ (2)

7.2



(Correct NOR symbol: OR shape with a circle at the output 1; truth table below 4)

(5)

A	B	Q
0	0	1
0	1	0
1	0	0
1	1	0

7.3 Q = A(1 + B) ✓ = A · 1 = A ✓; absorption law ✓ (3)

7.4 Any FOUR in order ✓✓✓✓: switch off and disconnect the power; heat the joint and remove the solder with a desoldering pump or braid; remove the faulty component, noting its orientation; clean the pads; fit the new component the correct way round; solder the joints neatly and cut off extra leads; inspect and test the circuit. (4)

7.5 Any TWO ✓✓: joint not heated enough (iron too cold or removed too soon); dirty or oxidised pad / lead; the joint moved while cooling; too little solder or flux. (2)

[20]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE